AliMark: Enhancing Robustness of Sentence-Level Watermarking Against Text Paraphrasing

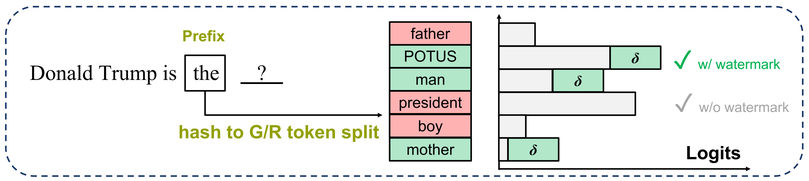
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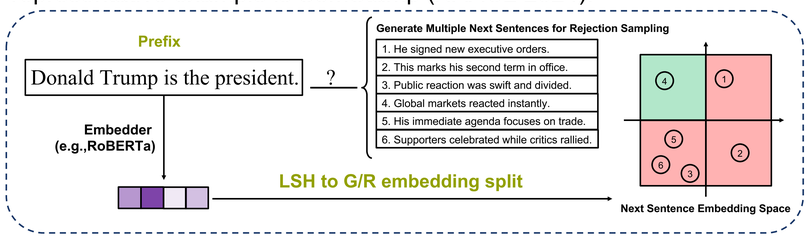
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Background: LLM Watermarking

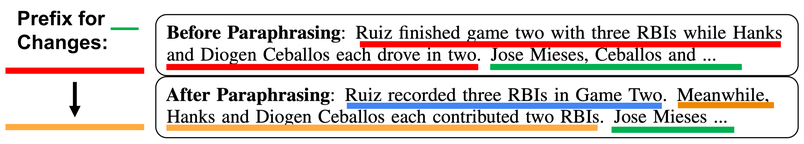
LLM watermark is an auditing method to verify whether a given text is generated by a specific LLM. A representative work, KGW (*ICML* 2023), embeds statistically detectable signals during next token generation.



However, token-level watermarks remain fragile to paraphrasing, as token substitutions directly weaken watermark signals. To address this limitation, sentence-level watermarking has been proposed, which anchors watermark signals in semantic space to improve robustness. A representative example is SemStamp (*NAACL* 2024).

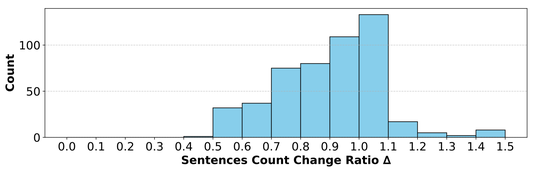


Structural Perturbations in Text-to-Text Paraphrasing



1 Real-world paraphrasers often introduce ***structural perturbations*** by merging consecutive sentences or splitting existing ones based on their logical structure. This behavior is particularly common in strong paraphrasers such as GPT-3.5. Below is empirical evidence:

$$\Delta = \frac{\#\text{sents after}}{\#\text{sents before}}$$



2 Such a prevalence of structural perturbations can substantially degrade watermark signals embedded by prefix-based sentence-level watermarking such as SemStamp (*NAACL* 2024), k-SemStamp (*ACL* 2024), and PMark (*ICLR* 2026), hindering their practical applicability.

Proposed Method: AliMark

A reformulation of sentence-level watermarking as bit sequence encoding and alignment tasks.

Watermark Embedding

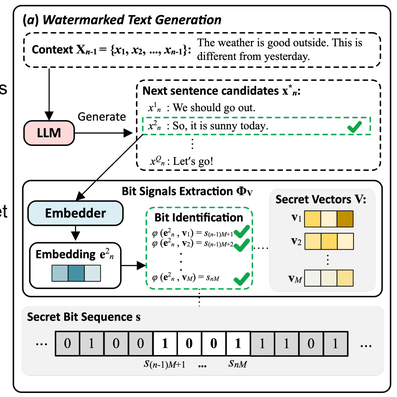
1 AliMark neither encodes nor detects watermark signals based on the prefix content. Instead, these operations are performed in a global manner.

2 It maintains a global secret bit sequence $\mathbf{s}$, which is partitioned into many blocks, each containing M bits.

3 During watermarked text generation, the n-th sentence will be selected from a pool of next sentence candidates, and the candidate, whose extracted bit signals match the n-th block of the secret bit sequence $\mathbf{s}$, is chosen.

$$b^q_{(n-1)M+m} \leftarrow \varphi(\mathbf{e}^q_n, \mathbf{v}_m) = \begin{cases} 0, & \text{if } \langle \mathbf{e}^q_n, \mathbf{v}_m \rangle < 0 \\ 1, & \text{otherwise} \end{cases}$$

4 This allows the entire text to encode a sequential watermark (bit signals), thereby enabling global-level alignment for the subsequent watermark detection.

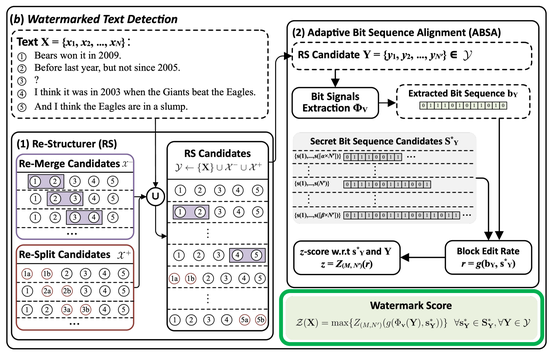


Watermark Detection

1 Re-Structurer (RS)

Main Goal: Attempt to re-construct the original watermarked text structure to enhance watermark signals.

- Two RS strategies are adopted: (1) re-merging two consecutive sentences, and (2) re-splitting an existing sentence at its midpoint. They are intended to address sentence splitting and merging operations caused by paraphrasers, respectively.
- These re-structuring operations are single-step, with all the resulting re-structured texts and the original input text constituting the RS candidates.



2 Adaptive Bit Sequence Alignment (ABSA)

Main Goal: Find the minimum alignment cost between the RS candidates and the global secret bit sequence $\mathbf{s}$.

Design Rationale: RS is insufficient for other forms of perturbations. For example, if an adversary deletes several sentences from a watermarked text, the subsequent re-merging and re-splitting operations are unlikely to reconstruct the original watermarked structure, necessitating additional mitigation strategies.

I. Secret Bit Sequence Candidates

- Due to potential adversarial insertions or deletions of sentences, the number of blocks in the ground-truth secret bit sequence corresponding to the original watermarked text may deviate from N', the number of blocks in a RS candidate.
- Hence, we construct a set of secret bit sequence candidates with varying numbers of blocks centered around N', with its range controlled by $\alpha < 1$ and $\beta > 1$:

$$\mathbf{S}^*_\mathbf{Y} \leftarrow \{\{\mathbf{s}(1), \cdots, \mathbf{s}(\lceil \alpha \times N' \rceil)\}, \cdots, \{\mathbf{s}(1), \cdots, \mathbf{s}(N')\}, \cdots, \{\mathbf{s}(1), \cdots, \mathbf{s}(\lceil \beta \times N' \rceil)\}\}$$

II. Block Edit Rate (BER)

An adaption of Levenshtein Distance to sentence-level watermark scenario to evaluate alignment cost:

- Block insertion and deletion have a fixed cost of M.
- Block substitution cost is the hamming distance.

$$D[i,j] = \min \begin{cases} D[i-1,j]+M \\ D[i,j-1]+M \\ D[i-1,j-1]+h(\mathbf{b}_\mathbf{Y}(i), \mathbf{s}^*_\mathbf{Y}(j)) \end{cases}$$

- The BER r is defined as the BED normalized by the max block size of the two input bit sequences.

III. Z-Score Calculation

- The BER is compared against the BER mean and standard deviation between two random bit sequences to obtain the watermark likelihood.

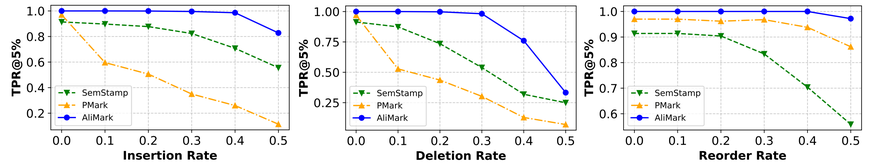
$$Z_{(M,N')}(r) = \frac{\bar{\mu}_{(M,N')} - r}{\bar{\sigma}_{(M,N')}}$$

Key Findings

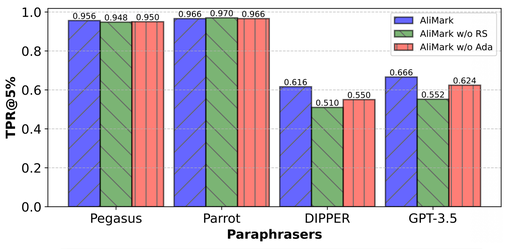
1 AliMark outperforms state-of-the-art token-level and sentence-level watermarking baselines under both weak and strong machine paraphrasing attacks in the majority of cases.

Dataset	Method	No Attack			Pegasus			Parrot			DIPPER			GPT-3.5		
		AUROC	TPR@1%	TPR@5%	AUROC	TPR@1%	TPR@5%	AUROC	TPR@1%	TPR@5%	AUROC	TPR@1%	TPR@5%	AUROC	TPR@1%	TPR@5%
Booksum	KGW	100.0	100.0	100.0	55.2	1.0	1.6	53.6	0.4	0.6	49.8	0.2	0.2	46.4	0.0	0.0
	EXP	99.2	49.1	98.8	87.7	4.8	41.4	86.6	7.9	44.3	78.4	2.4	21.1	62.2	0.2	2.4
	SynthID	100.0	100.0	100.0	90.5	5.8	26.4	93.5	9.8	37.2	70.9	0.0	1.8	62.6	0.0	1.0
	SIR	99.4	14.6	93.6	35.8	0.0	0.4	28.0	0.0	0.0	35.9	0.0	0.6	32.9	0.0	0.2
	SemStamp	98.3	91.4	91.4	94.0	65.8	70.8	96.1	67.9	73.6	81.6	20.3	26.7	78.0	17.0	22.0
	k-SemStamp	100.0	98.2	98.5	96.2	60.5	62.8	97.3	68.5	70.2	78.4	12.7	14.1	78.1	15.3	16.6
	SimMark	99.5	70.8	91.0	99.0	45.8	76.2	99.1	43.6	78.0	87.9	6.4	23.4	85.0	6.6	18.2
	PMark	99.5	97.0	97.0	97.2	82.6	86.0	99.3	92.0	93.0	86.5	26.6	30.4	88.6	29.2	33.0
	AliMark	100.0	100.0	100.0	**99.8**	**93.8**	**95.6**	**94.2**	**55.8**	**61.6**	**96.1**	**60.2**	**66.6**			
C4	KGW	100.0	100.0	100.0	50.0	1.0	1.0	54.1	0.0	0.2	53.5	0.6	0.6	46.5	0.2	0.4
	EXP	99.8	99.6	99.6	88.2	47.5	51.9	89.2	51.1	54.3	76.7	21.0	24.8	64.6	6.2	8.2
	SynthID	100.0	100.0	100.0	92.7	36.8	46.6	91.6	43.6	51.4	70.9	5.8	8.4	66.1	3.4	5.4
	SIR	99.9	95.4	98.2	45.6	0.2	1.2	37.6	0.0	0.2	42.2	0.0	0.2	44.9	0.0	0.4
	SemStamp	97.8	84.2	84.9	92.7	47.1	53.5	93.0	53.4	56.9	79.1	14.7	17.9	75.2	12.7	16.8
	k-SemStamp	99.3	23.3	91.1	93.5	2.4	33.9	94.8	3.0	39.9	81.5	0.2	9.0	77.6	0.3	6.1
	SimMark	98.6	13.8	75.4	97.4	6.6	58.2	97.7	4.0	60.6	89.1	1.2	20.2	80.8	0.6	8.6
	PMark	98.9	94.8	95.0	96.1	**78.6**	79.6	98.1	**88.6**	89.4	84.2	28.6	29.6	85.1	26.4	28.2
	AliMark	99.9	99.2	99.4	**98.2**	69.6	**81.2**	**98.9**	83.6	**91.2**	**91.5**	**37.0**	**49.8**	**92.3**	**36.0**	**51.6**

2 AliMark also exhibits stronger robustness under probing attacks, which only alter the text structure, including the random insertion of sentences and the random deletion or reordering of watermarked sentences.



3 Each detection module in AliMark makes a meaningful contribution, with performance improvements becoming particularly pronounced when evaluated against stronger paraphrasers. The detection runtime overhead from the RS module is a reasonable trade-off for the enhanced robustness.



Method	Time (s)					
	$N = 4$	$N = 8$	$N = 16$	$N = 32$	$N = 64$	$N = 128$
AliMark	0.02	0.03	0.05	0.08	0.15	0.34
AliMark w/o RS	0.02	0.02	0.02	0.03	0.05	0.07
AliMark w/o Ada	0.02	0.03	0.04	0.07	0.14	0.27

Resource

Paper

Code

Feel free to connect!

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